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| **Fiscal Year:** | FY 2016 | **Task Last Updated:** | FY 04/08/2016 |
| **PI Name:** | Bailey, Susan M. Ph.D. | | |
| **Project Title:** | Differential Effects on Telomeres and Telomerase in Twin Astronauts Associated with Spaceflight | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Biomedical countermeasures | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
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| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | susan.bailey@colostate.edu | **Fax:** | FY 970-491-7742 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 970-491-2944 |
| **Organization Name:** | Colorado State University | | |
| **PI Address 1:** | Environmental and Radiological Health Sciences | | |
| **PI Address 2:** | 1618 Campus Delivery | | |
| **PI Web Page:** | | | |
| **City:** | Fort Collins | **State:** | CO |
| **Zip Code:** | 80523-1618 | **Congressional District:** | 2 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2013-14 HERO NNJ13ZSA002N-TWINS |
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| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Charles, John | **Contact Phone:** | 281-483-7224 |
| **Contact Email:** | jbcharle@gmail.com | | |
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| **Task Description:** | The ends of human chromosomes are capped by telomeres, tandem arrays of repetitive DNA sequence, which together with associated proteins serve to protect chromosomal termini from inappropriate degradation. Functional telomeres also prevent these natural DNA ends from being recognized as broken DNA (double-strand breaks--DSBs) and triggering inappropriate DNA damage responses, which could result in chromosomal instability.<br>It is well established that telomere length diminishes with cellular division and normal aging. It is also becoming increasingly appreciated that telomere length is influenced by a variety of other factors including life stress, infection, inflammation, and oxidative stress. Nobel Laureate Elizabeth Blackburn suggests that telomere shortening "powerfully quantifies life's insults." Indeed, recent research supports telomere length maintenance as a key integrating component for the cumulative effects of genetic, environmental, and lifestyle factors on aging and aging-related diseases. Thus, the rate at which telomeres shorten provides an informative biomarker of biological aging and further, erosion of telomere length can be linked to age-related pathologies including reduced immune function, loss of fertility, cardiovascular disease, and cancer.<br><br>Healthy lifestyles and diets are positively correlated with telomere length, and changes in diet and lifestyle have also been shown to modulate telomerase activity. Telomerase is the specialized enzyme present in germ line, stem, and cancer cells, capable of de novo addition of telomeric repeats onto the ends of newly synthesized telomeres. A large remaining puzzle in human populations is the relationship between telomerase activity (e.g., measured in peripheral blood lymphocytes) and lifestyle factors (e.g., nutritional, psychological, and physical stressors). We speculate that for the astronauts, telomere length represents a particularly relevant biomarker, because it reflects the combined exposures and experiences encountered during space travel. That is, an individual astronaut's genetic susceptibilities, exposures to galactic cosmic radiation, physical, and psychological stressors encountered, are all "captured" as changes in telomere length.<br><br>A remarkable once-in-a-life time opportunity currently exits to evaluate the effects of spaceflight factors on the monozygotic twin astronauts Scott and Mark Kelly, who will have spent a total of 540 and 54 days in low Earth orbit, respectively. We propose to assess differential effects on telomere length and telomerase activity associated with the upcoming yearlong International Space Station (ISS) mission in the twins – individuals matched genetically and who share similar, unique lifestyles; i.e., they are of similar "nature and nurture." Standard assays will be employed to determine telomere length and telomerase activity as part of their respective and concurrent biodosimetry/chromosome analyses pre-flight (to establish baselines), during flight (to evaluate short-term/temporary changes), and post-flight (to evaluate long-term/permanent changes). Results from the space- and ground-based twins will be compared to evaluate effects of spaceflight factors on these informative biomarkers. Results from the twins will also be compared with our current retrospective analyses of a cohort of unrelated astronauts who have participated in missions of 3 to 6 months duration to evaluate inter-individual sensitivity to these effects of spaceflight, a strategy that will also facilitate evaluation of time dependent responses. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Aging and age-related degenerative pathologies like dementia, cardiovascular disease, and cancer are everyday concerns on Earth as well as in space. Taken together with the other TWINS Study investigations, results from this research will facilitate identification of interactive effects and relationships that will suggest potential mitigation strategies for future study, as well as benefit precision medicine efforts in monitoring general health, and assessing aging and disease risk not only of astronauts, but for those on Earth as well. |
| **Task Progress:** | This study is part of an integrated compilation of multiple studies led by multiple Principal Investigators (PIs) at multiple centers that takes advantage of a remarkable and unique opportunity to investigate the effects of space travel with monozygotic twin astronauts, who will have spent a total of 540 and 54 days in low Earth orbit; importantly, they are of similar "nature and nurture." For the purposes of the Twins study, the flight subject will be on-orbit (ISS) for one year, while the ground subject will remain Earth-bound for that same year. This particular project will interrogate changes in telomere length and telomerase activity associated with prolonged and combined exposures to galactic space radiations, microgravity, and other spaceflight stressors (e.g., nutritional, psychological, physical). Samples from each of the twins will be assessed for telomere length and telomerase activity at similar times before the flight, during flight, and after return to Earth, to determine whether observed changes differ relative to each other (i.e., are space-flight specific), as well as how they relate to physiological, molecular, genetic, and biochemical changes (information obtained via data sharing with other twins studies).<br>Telomere length maintenance captures the interplay between genetics and life experiences/exposures, and so is a key integrating component for the cumulative effects of genetic, environmental, and lifestyle factors on aging and age related degenerative diseases, ranging from reduced immune function, to cardiovascular disease and cancer. Therefore, we propose that for the astronauts, telomere length maintenance represents a particularly relevant biomarker, as it reflects the combined exposures and experiences encountered during space travel. Our overall goal is to test the hypothesis that accelerated telomere shortening and elevated telomerase activity will be associated with space flight. Progress on Specific Aim 1 continues:<br><br>Aim 1. Assess differential effects on telomere length and telomerase activity concurrently in the space- and ground-based twin astronauts, and in ground-based age-matched controls.<br><br>• Blood samples will be taken pre-flight (to establish baseline), during flight (to evaluate short-term/temporary changes) and post-flight (to evaluate long-term/permanent changes).<br><br>• Comparisons between the twins controls variables of individual genetic differences, susceptibilities and lifestyle factors as much as possible, making observed differences space-flight specific.<br><br>• Comparisons with and between unrelated astronauts and age-matched controls (separate study) will allow evaluating role of genetics and individual susceptibilities (on-going prospective & retrospective study of a cohort of unrelated astronauts who have participated in missions of 3-6 months duration, which includes the flight subjects previous flight).<br><br>• Coordinated with previous and on-going biodosimetry studies; chromosome aberration analyses are also being conducted to evaluate specific contribution of ionizing radiation exposure.<br><br>In association with the one-year mission, blood samples have been taken pre-flight (to establish baseline), in-flight (to |

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| | evaluate short-term/temporary changes), and will continue post-flight (to evaluate long-term/permanent changes). Telomere length is being assessed on a cell-by-cell basis using TELO-FISH; Florescence in situ Hybridization (FISH) with a telomere probe on chromosomes (and interphase nuclei), which is evaluated as Relative Fluorescence Intensity (RFI) and results in distributions of telomere length (shortest to longest). Telomerase activity is being assessed using the quantitative Real Time-PCR Telomere Repeat Amplification Protocol (qRT-PCR TRAP).<br><br>Deidentified pre-flight (2), in-flight (2), and first post-flight (1) samples on both the in-flight and ground subjects (and two ground-based age-matched control subjects) have been successfully collected and analyses are ongoing. We are currently in the process of collecting remaining post-flight samples (3) from both the in-flight and ground subjects (and two ground-based age-matched control subjects) and processing for analyses.<br><br>Once all post-flight samples have been collected, and to the extent possible, all samples will be run as a single batch to control for potential batch effects. Taken together with the other Twins study investigations (data sharing), results will facilitate identification of interactive effects and relationships that will suggest potential mitigation strategies for future study, as well as benefit precision medicine efforts in monitoring health and assessing disease risk not only of astronauts, but for those on Earth as well. |
| **Bibliography Type:** | Description: (Last Updated: 12/10/2025) |
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| **Abstracts for Journals and Proceedings** | Bailey SM. "Telomeres, Stress & Aging." Platform presentation at Front Range Consortium on Stress Resistance and Aging, Fort Collins, CO, Oct 30, 2015.<br>Front Range Consortium on Stress Resistance and Aging, Fort Collins, CO, October 30, 2015. , Oct-2015 |
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| **Abstracts for Journals and Proceedings** | Sishc BJ, Nelson CB, McKenna MJ, Battaglia CLR, Herndon A, Idate R, Liber HL, Bailey SM. "Telomeres and Telomerase in the Radiation Response." Platform presentation at Space Radiation and Heavy Ions in Therapy Symposium 2015, Osaka, Japan, May 22-24, 2015.<br>Space Radiation and Heavy Ions in Therapy Symposium 2015, Osaka, Japan, May 22-24, 2015. , May-2015 |
| **Abstracts for Journals and Proceedings** | Bailey SM, Sishc BJ, Nelson CB, McKenna MJ, Taylor L, George K. "Assessing Telomere Length and Telomerase Activity in Astronauts." Poster presentation at the 15th International Congress of Radiation Research, Kyoto, Japan, May 25-29, 2015.<br>15th International Congress of Radiation Research, Kyoto, Japan, May 25-29, 2015. , May-2015 |
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